Martim Afonso

PhD Candidate in Computer Science · Formal Verification & Agentic Theorem Proving

martim.afonso@tecnico.ulisboa.pt | Lisbon, Portugal | timafonso.github.io | in.com/in/martim-afonso-79b1631a2/ | https://github.com/timafonso



Profile

PhD candidate in Computer Science at the University of Lisbon (INESC-ID / Instituto Superior Técnico), working on automating the formal verification of software through Agentic Theorem Proving.

Education

Ph.D. in Computer Science 2023 – Present

University of Lisbon — INESC-ID / Instituto Superior Técnico · Lisbon, Portugal

- Automating the formal verification of software through Agentic Theorem Proving.
- Driving tools such as Lean and Verus with language models and machine learning.

M.Sc. in Computer Science 2021 – 2023

Instituto Superior Técnico, University of Lisbon · Lisbon, Portugal

- Thesis: Finding Interesting Regions in Whole Slide Images Using Multiple Instance Learning

B.Sc. in Computer Science 2018 – 2021

Instituto Superior Técnico, University of Lisbon · Lisbon, Portugal

Experience

Visiting Student Sep – Dec 2024

Carnegie Mellon University · Pittsburgh, USA

- Worked on learning-based fault localization for Python codebases.

Backend Engineer Intern Aug - Oct 2022

Sky · Lisbon, Portugal

- Developed a tool for automated project template generation, including the infrastructure-as-code files needed for testing and deployment.

Publications

ProfOlaf: Semi-Automated Tool for Systematic Literature Reviews 2026

34th ACM Int. Conf. on the Foundations of Software Engineering (FSE)

ProofCoder: Continuous Reasoning for Automated Software Verification 2026

arXiv preprint 2611.14106

SeVerus: Context-Aware, Gamified Benchmarking for Rust Verification 2026

arXiv preprint 2611.07194

VerusWeave: A Benchmark for Proof Repair in Concurrent Verus Programs 2026

arXiv preprint 2611.00287

Multiple Instance Learning for WSI: A comparative analysis of attention-based approaches 2024

Research Interests

- Automated Theorem Proving (Lean, Rocq)
- Formal verification of software (Verus)
- Machine Learning & Language Models
- Explainable AI for agent-assisted verification

Languages

- Portuguese (native)
- English (fluent)

Profiles

- Google Scholar: scholar.google.com/citations?user=HmLTs0AAAAAJ
- ORCID: 0009-0002-9271-9276